

RHIZARIANS

The clade Rhizaria includes two phyla that are among the most beautiful of all of the miniature protocists: the radiolarians and the forams.

The unique, intricately sculptured shells of both phyla make them distinctive members of the microscopic world, and some have left an impressive fossil record. Most radiolarians build a glassy shell with silica, stripping the ocean surface of this otherwise abundant mineral when they bloom in large numbers. Their body radiates long pseudopods that poke through their shell like the rays of the sun, supplemented in some by hardened spines. Radiolarians use their pseudopods for catching food, but some also harbour living algae, gaining nourishment from the sugars their

photosynthetic partners make as they drift in sunlit tropical oceans. Their silica dependence binds radiolarians to the sea, but their allies, the cercozoans, have also exploited soil and fresh water. Cercozoans typically retain the long pseudopods, but have shelled, and shell-less forms, and some have flagella, depending on habitat demands.

The forams, or foraminiferans, have flourished in the oceans for hundreds of millions of years and their calcium carbonate shells litter the seabed to form layers of chalky sediment. Forams shells are so distinctive, even when fossilized, that geologists can use them to date deposits and identify hidden reservoirs of oil. When alive, each shell harbours a tiny amoeba that hunts with pseudopods – like the radiolarians – although some are big enough to catch animal larvae.

DOMAIN	EUKARYOTA
KINGDOM	PROTOCTISTA
CLADE	RHIZARIA
FAMILIES	108
SPECIES	About 14,000

DEBATE CONVERGING GIANTS

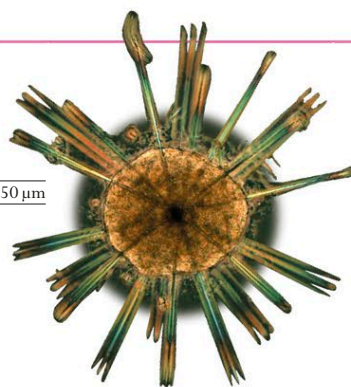
The pseudopods of radiolarians and forams enmesh to form nets around their cells. This perhaps supports the idea that both forms descend from a common ancestor. But it is possible that this net evolved independently in the two groups, perhaps a result of both having large single cells.

LATTICE-SHELLED RADIOLARIAN

This radiolarian species of plankton belongs to the Spumellaria order, most of which have silica latticework skeletons.



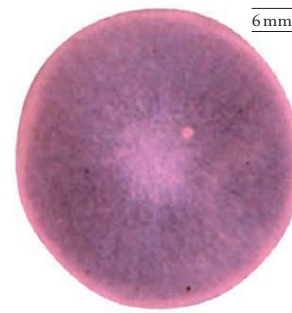
100 µm



350 µm

STAR RADIOLARIAN *Astrolithium* sp.

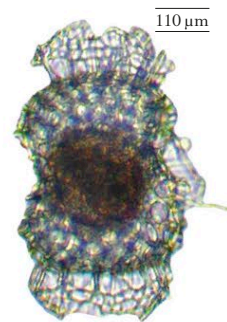
In this radiolarian, the bases of the radial spines are fused into a star-shaped mass, or central body.



6 mm / ¼ in

SPHERICAL RADIOLARIAN *Physematium* sp.

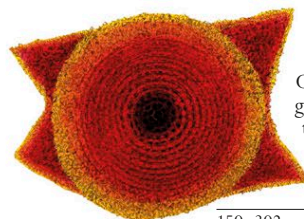
The cell surface of this species holds photosynthesizing organisms in a mesh of thin pseudopods.



110 µm

DIDYMOCYSTIS TETRATHALAMUS

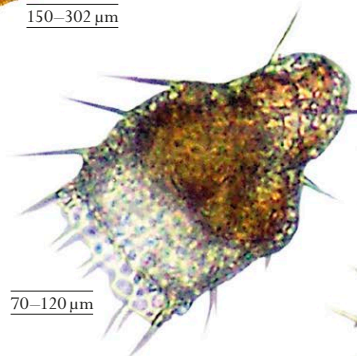
A living representative of an ancient fossil lineage, this radiolarian is named for its twin shells, which sometimes have hemispherical or conical caps.



150–302 µm

SPONGASTERICUS QUADRICORNIS

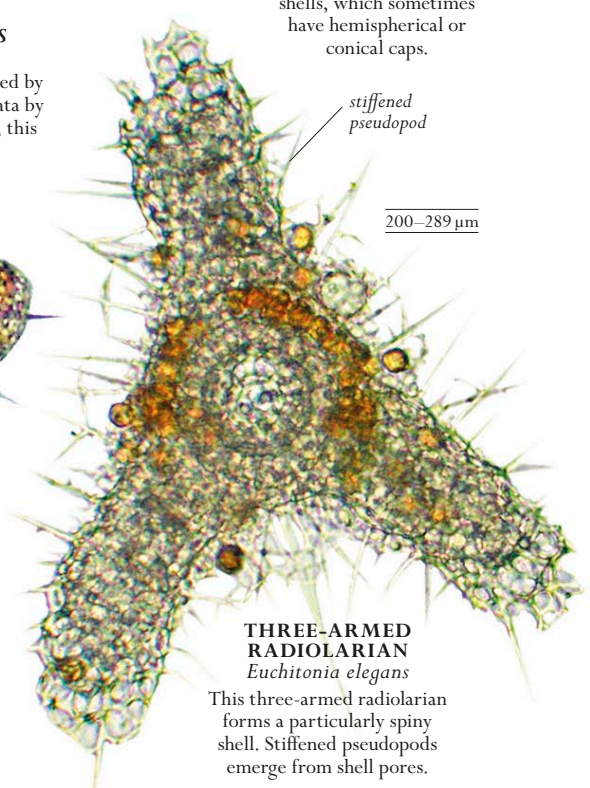
One of the radiolarians used by geologists to help date strata by the presence of its fossils, this species' name means "four-cornered".



70–120 µm

SPINY RADIOLARIAN *Lithomelissa setosa*

This species displays the greatest seasonal variation of any radiolarian, making it a useful seasonal indicator in fossil and sedimentary records.



stiffened pseudopod

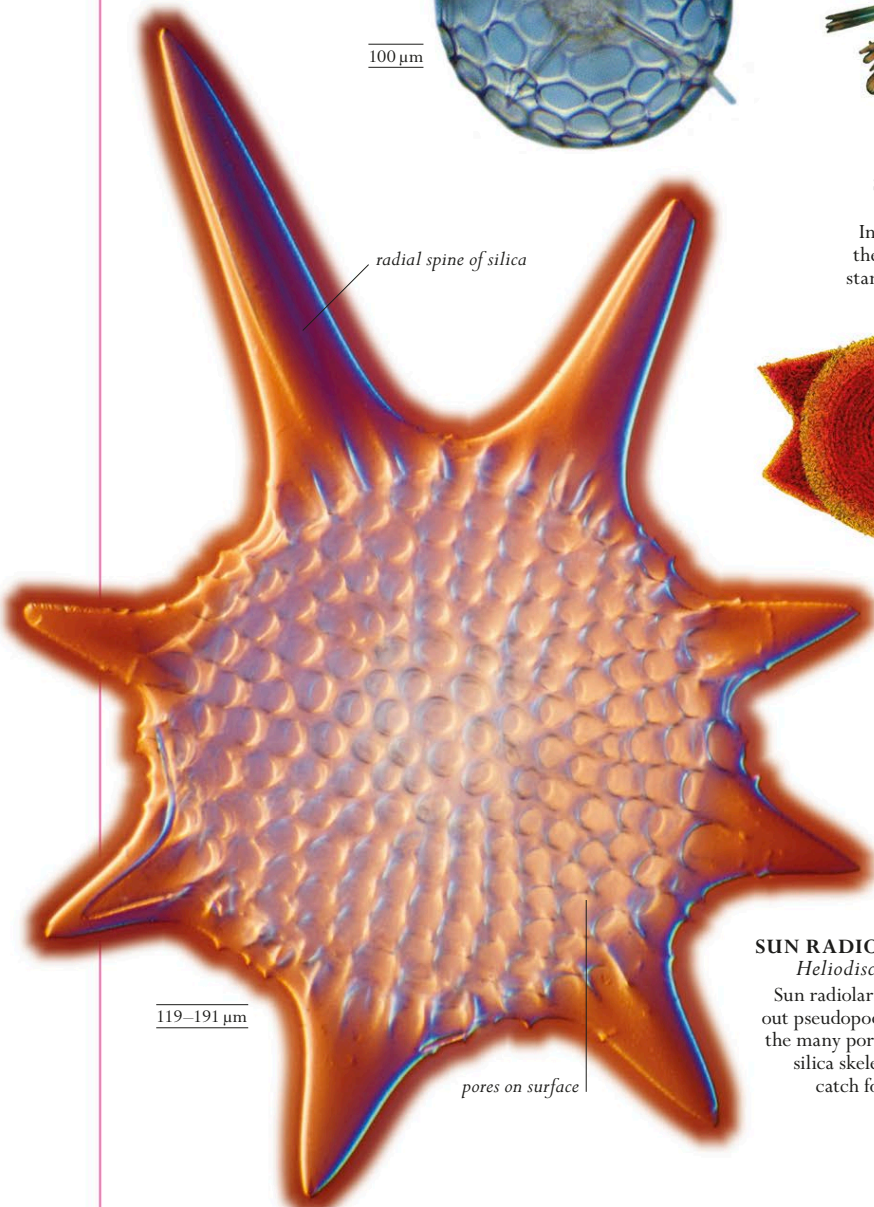
200–289 µm

THREE-ARMED RADIOLARIAN *Euchitonina elegans*

This three-armed radiolarian forms a particularly spiny shell. Stiffened pseudopods emerge from shell pores.

SUN RADIOLARIAN *Heliodiscus* sp.

Sun radiolarians send out pseudopods through the many pores in their silica skeleton to catch food.



radial spine of silica

pores on surface

119–191 µm